

Telco Customer Churn Analysis

1. Project Objective

Business Problem

The telecom company is experiencing customer churn. The objective of this analysis is to understand the scale of churn, identify customer segments with higher churn rates, and determine the major reasons customers leave.

Key Questions

1. How many customers have churned?
 2. What is the overall churn rate?
 3. Which customer groups have higher churn rates?
 4. Which contract and internet service types have higher churn?
 5. What are the major reasons customers leave?
 6. How does tenure relate to customer churn?
 7. Are monthly charges different between churned and retained customers?
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2. Database & Table Creation

Database

Created a MySQL database named Churn.

Main Table

Created **telco_churn** containing customer demographics, services, contract information, billing information, and churn information.

Main data areas:

- Customer information
 - Demographics
 - Service information
 - Contract information
 - Payment information
 - Charges
 - Churn information
 - Churn reasons
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3. Data Transformation

Two additional columns were created to support analysis.

Churn_Reason_Category

Grouped detailed churn reasons into broader business categories:

Category	Purpose
Competitor	Customers leaving because of competitor offers/services
Service	Service and support-related issues
Pricing	Price and additional charge-related issues
Network	Network and speed-related issues
Product	Product/service offering dissatisfaction
Digital Experience	Website/self-service issues
Life Change	Customer circumstances such as moving
Unknown	Customers where the reason is unknown

For customers where Churn_Label = 'No', the category is kept as NULL.

Tenure_groups

Customers were divided into:

- 0–12 months
- 13–24 months
- 25–48 months
- 49+ months

This was created to compare churn across different stages of the customer lifecycle.

4. Exploratory Data Analysis

Then document your SQL analysis by **business question**, not by SQL query number.

4.1 Churn Overview

Question: How large is the churn problem?

SQL Analysis:

- Total churned customers: - **1869**
- Overall churn rate: - **26.58%**
- Average tenure of churned customers: - **18** months

Finding:

1,869 customers churned, representing approximately **26.58%** of the customer base.

4.2 Churn by Gender

Question: Does churn differ between male and female customers?

Analysis:

- Total customers by gender: - male = **3549** | female = **3483**
- Churned customers by gender: - male = **930** | female = **939**
- Churn rate by gender: - male = **26.20%** | female = **26.96%**

Churn rates were similar across gender, with **26.20%** for **males** and **26.96%** for **females**.

4.3 Churn by Internet Service

Question: Which internet service has a higher churn rate?

Analysis:

- DSL: - Total = 2416 | Churned = **459** | Percentage = **19%**
- Fiber Optic: - Total = 3096 | Churned = **1297** | Percentage = **41.89%**
- No: - Total = 1520 | Churned = **113** | Percentage = **7.43%**

Finding:

Fiber optic customers had an observed churn rate of approximately **41.89%**, compared with **19%** for **DSL** customers.

Then add:

This is an observed association in the dataset and does not establish that **Fiber optic** service causes churn.

4.4 Churn by Contract Type

Question: How does contract type relate to churn?

Analysis:

- Month-to-month: - Total = 3875 | Churned = **1655** | Percentage = **42.71%**
- One year: - Total = 1472 | Churned = **166** | Percentage = **11.28%**
- Two years: - Total = 1685 | Churned = **48** | Percentage = **2.85%**

Average Tenure:

- Month-to-month: - **18**
- One year: - **42**
- Two years: - **57**

It is clearly seen as month-to-month contract has more churn and least avg tenure.

4.5 Churn by Payment Method

Question: Does churn vary across payment methods?

Analysis:

- Mailed check: - Total = 1604 | Churned = **308** | Percentage = **19.20%**
- Electronic check: - Total = 2365 | Churned = **1071** | Percentage = **45.29%**
- Credit card (automatic): - Total = 1521 | Churned = **232** | Percentage = **15.25%**
- Bank transfer (automatic): - Total = 1542 | Churned = **258** | Percentage = **16.73%**

Electronic Check customers have a **45.29%** churn rate, which is higher than other payment methods.

4.6 Monthly Charges

Question: Do churned customers have different monthly charges from retained customers?

Analysis:

- Average monthly charges — churned = **\$74.44**
- Average monthly charges — retained = **\$61.31**
- Difference = **\$13.13**

Churned customers had higher average monthly charges (**\$74.44**) than retained customers (**\$61.31**), a difference of **\$13.13**.

4.7 Churn Reasons

Question: Why are customers leaving?

Analysis:

- Competitor: - **621**
- Support: - **231**
- Service: - **224**
- Pricing: - **199**
- Unknown: - **154**
- Network: - **147**
- Product: - **146**
- Digital Experience: - **88**
- Life Change: - **59**

Most churn occurred due to **good deals given to customers** by **competitor** and **due to bad support** and **service**.

4.8 Churn by Tenure Group

Question: Which tenure group has the highest churn rate?

Document:

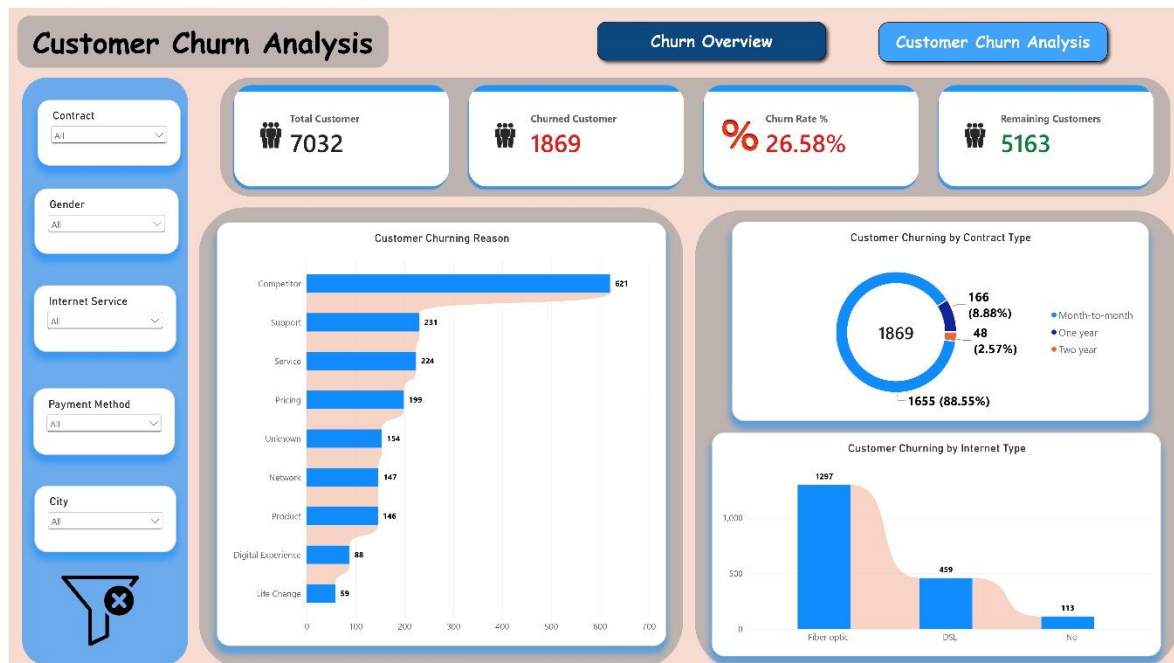
Tenure Group	Customers	Churned	Churn Rate
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0–12	2175	1037	47.68%
13–24	1024	294	28.71%
25–48	1594	325	20.39%
49+	2239	213	9.51%

Customers with **0–12** months of tenure have the highest churn rate at **47.68%**, while customers with **49+** months have the lowest at **9.51%**.

5. Power BI Dashboard

The SQL analysis was used to prepare the dataset for Power BI visualization. The dashboard focuses on customer churn, contract type, internet service, payment method, tenure, and churn reasons.



6. Key Findings

1. **26.58%** of customers churned, indicating a significant customer-retention issue in the dataset.
2. **Fiber optic** customers have a higher churn rate than **DSL** customers, making Fiber customers an important segment for further investigation.
3. **Contract type shows a strong difference in churn**, with month-to-month customers experiencing higher churn compared with one-year and two-year contract customers.
4. **45.29% of customers use Electronic Check as their payment method**, making it the most commonly used payment method in the dataset.
5. **33.23% of churned customers had competitor-related reasons**, including better offers, higher speeds, more data, and better devices.
6. **Support and service-related reasons account for 24.35% of recorded churn reasons**, with 12.36% related to support and 11.99% related to service.
7. **Pricing-related issues also appear among the recorded churn reasons**, including high prices, extra data charges, and long-distance charges.

7. Business Recommendations

1. Improve Early Customer Retention

A large share of churn comes from customers with shorter tenure. The company should investigate early-stage customer experience and strengthen onboarding, guidance, and support during the initial months.

2. Investigate Fiber Optic Customer Churn

Fiber optic customers show a higher observed churn rate than DSL customers. The company should investigate potential issues related to service quality, network reliability, speed, and customer expectations.

3. Encourage Longer-Term Contracts

Month-to-month customers show higher churn compared with customers on longer-term contracts. The company could explore incentives, discounts, or additional benefits to encourage customers to move toward one-year or two-year contracts.

4. Address Competitor-Related Churn

Competitor-related reasons represent a major portion of recorded churn. The company should regularly review competitor offerings, including pricing, data allowances, internet speeds, and device options, to identify areas where its offering may be less attractive.

5. Improve Service and Support Experience

Support and service-related issues appear frequently among churn reasons. The company should investigate support interactions, response quality, and employee expertise to identify areas for improving the customer experience.

6. Review Pricing and Additional Charges

Pricing-related reasons are also present among churned customers. The company should review pricing structures, additional charges, and billing practices to identify potential sources of customer dissatisfaction.
